

Counting Bishops

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to count the number of ways k bishops can be placed on an $n \times n$ chessboard so that no two bishops attack each other.

Two bishops attack each other if they are on the same diagonal.

Input

The only input line has two integers n and k : the board size and the number of bishops.

Output

Print one integer: the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 500$
- $1 \leq k \leq n^2$

Example

Input	Output
5 4	2728